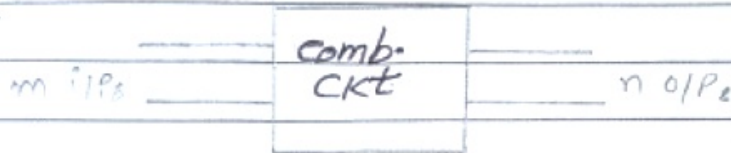
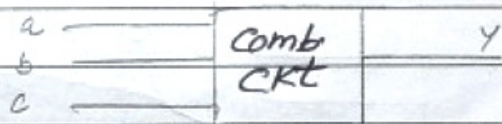


Design Procedure of combinational circuit.



- Step 1: Determine total no. of inputs and outputs.
- Step 2: Draw truth table with help of given statement.
- Step 3: With the help of boolean algebrae or Kmap, minimize every output which is function of inputs.
- Step 4: With the help of suitable logic gates, implement every o/p function.

Combinational circuits has 3 i/p's and 1 o/p.
o/p becomes high when majority no. of inputs are high, otherwise low.



a	b	c	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$Y = \sum m(3, 5, 6, 7).$$

0

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01

11

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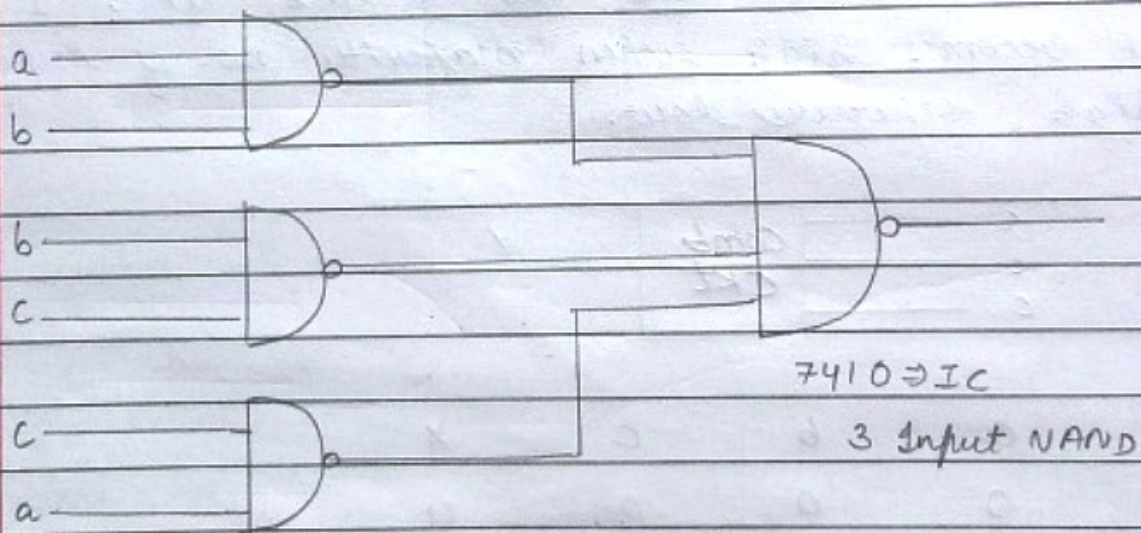
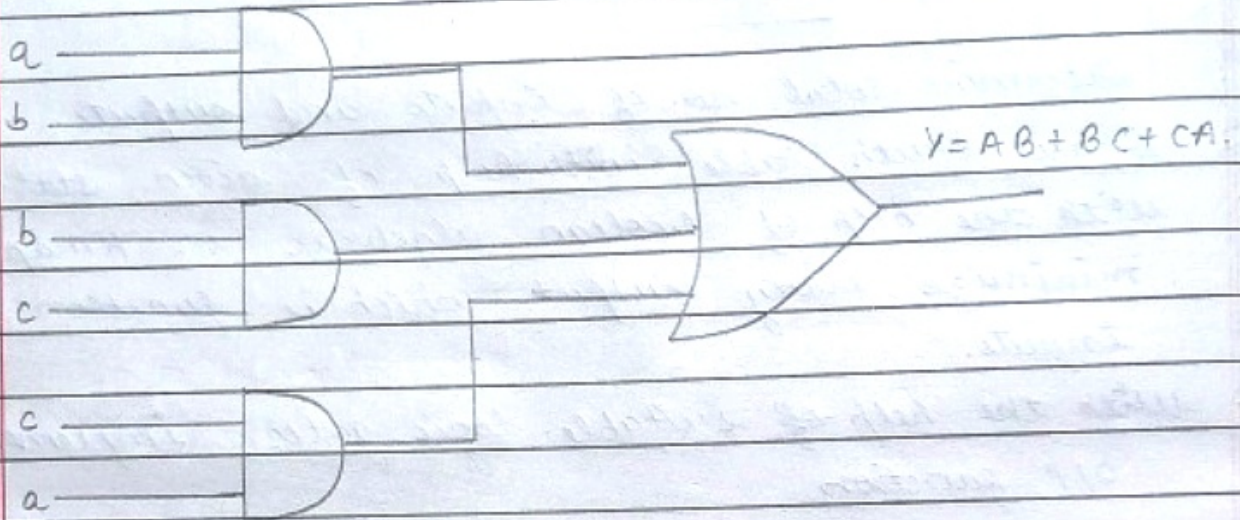
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$$Y = AB + BC + CA$$



Q Design a combinational circuit which has 4 i/p's and 1 o/p. and output becomes high when it's divisible by 5, else low.

	a	b	c	d	O/P.
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	0
10	1	0	1	0	1
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	0
15	1	1	1	1	1

AB \ CD	CD			
	00	01	11	10
$\bar{A}\bar{B}$	00	1		
$\bar{A}B$	01		1	
AB	11		1	
$A\bar{B}$	10			1

$$\begin{aligned}
 y &= \bar{a}\bar{b}\bar{c}\bar{d} + \bar{a}b\bar{c}d + a\bar{b}cd + a\bar{b}c\bar{d} \\
 &= \bar{a}\bar{c}(bd + \bar{b}d) + ac(bd + \bar{b}d) \\
 &= (\bar{a}\bar{c} + ac)(bd + \bar{b}d) \\
 &= (a \odot c)(b \odot d)
 \end{aligned}$$